Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/17/2007
PI Name:	Theofanous, Theofanis G. Ph.D.		
Project Title:	The Limits of Coolability in Thermal Systems for Space		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	10/01/2001	**End Date:**	11/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 11/30/2006, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dinh, Truc-Nam (University of California Santa Barbara)		
Grant/Contract No.:	NAG3-2761		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project is an integrative effort addressing practical aspects of boiling in the HEDS context. The work is a continuation and extension of a project that addressed coolability limits in pool boiling carried out over the previous funding cycle. A key result of this previous work was the identification of a scales separation phenomenon, that effectively "isolates" the microhydrodynamics on the heater surface from the chaotic two-phase flow motions away from it. This separation is effected by a persistent very high void fraction region (like a vapor "blanket"), and it is of immense significance in allowing to focus the physics of a long-standing seemingly intractable problem. Experiments and numerical simulations in this project are used to examine applicability of scales separation as an organizing principle in convective boiling and boiling in fractional gravities. Task Objective: 1. Develop a basic understanding for the fine structure of the extended micro-layer, as revealed through our "scale separation" idea and the dynamic thermal patterns on a nano-film heater. 2. Define appropriate, measurable surface properties relevant to the rupture of micro-layer, and relate those (quantitatively) to fluid properties and the rupture condition (the critical heat flux). 3. Determine the extent of applicability of scales separation in reduced gravity; that is, the point where CHF becomes water-supply-limited. 4. Determine the influence of convection on the results of Objectives 1 and 2. 5. Translate the results of all previous objectives to terms usable for predicting and enhancing CHF, as well as for use in a two-phase CFD framework. This project aims to provide a basis to develop a physics-based understanding of Boiling Crisis phenomenon in pool boiling (horizontal and vertical flat plates), and weakly-convecting flow boiling (vertical channel flow). We can then translate this understanding to a means for quantitative prediction of critical heat flux, as a function of local (near-wall) two-phase flow conditions, appropriate and convenient for interrogation (modelling) in the frame of detailed multiphase CFD simulations. Furthermore, the thus-gained understanding and experience can be used to enhance the coolability limits in both pool and forced convection flows by manipulating surface characteristics, and perhaps even flow geometries.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Westinghouse Electric Company, together with the US Department of Energy leveraged resources with this NASA-funded work to address certain practical issues in the next generation large nuclear power reactors. In particular, the basic understanding derived from BETA experiments guided the design and burnout testing in a large-scale, flow-boiling facility ULPU-2400, which provided vital support for certification (by the US Nuclear Regulatory Commission) of Westinghouse severe accident management approach in their new, AP1000, reactor design. In a complementary fashion, resources made available through this comparatively large test program contributed significantly to the pursuit of the present fundamentally oriented, research program for NASA.
	The project was completed. In the project, a broad range of pool boiling and liquid-film boiling experiments were conducted to examine the effect of various physico-chemical factors (surface treatment, coolant chemistry, nanoparticles) on nucleation, boiling heat transfer and burnout. Using a high-speed video imaging and infrared thermometry technique, we identified nucleation-induced hot-spot expansion and rewetting as key mechanisms that govern the micro-hydrodynamics in the evaporating thin liquid film on a high heat-flux heated surface. The insights from experiments paved the way to modeling and numerical simulations, which supported the notion that under high heat (evaporation) flux, craters are formed in a thin liquid film and expand laterally to very large sizes due to the vapor recoil force exerted at the contact line. The basic understanding derived is fundamental in delineating the role of nucleation site density, their dynamic interactions, surface wettability, and liquid supply on burnout. Most notably, the novel physical picture serves as common dominator for burnout in pool boiling, flow boiling and spray cooling. The project focus is on mechanisms which govern burnout in pool boiling. The work capitalized on advanced experimental and diagnostic techniques that have been developed at UCSB, namely the use of high-speed high-resolution infrared thermometry on a nano-film heater vapor-deposited on a thin glass substrate (BETA experiments). In addition, high-speed video imaging was used together with appropriate lighting to visualize the microhydrodynamics of an evaporating thin liquid film (BETA-B configuration). Experimentally, the effect of water chemistry, surface treatment, and heater material on pool boiling CHF was investigated. Extensive measurements were performed with X-Ray radiography to further quantify the scale separation phenomenon discovered in our previous studies on the BETA facility. Theoretical research was directed toward understanding mechanisms that control heterogeneous nucleation, including that on nanoscopically smooth surfaces. The study indicated that both surface heterogeneity (Franken islands) and gas solubility in water may have contributed to the reduced nucleation energy barrier (Dinh et al, 2003b). Critical heat flux was measured for boiling on horizontal, upwards-facing copper and steel surfaces under the influence of various coolant chemistries. Solutions of Boric Acid (BA), TriSodium Phosphate (TSP), pure De-Ionized Water (DIW) and Tap Water (TW) were tested in experiments carried out in the BETA facility. We found that the presence of chemicals TSP and BA enhances the coolability of fresh surfaces well beyond even those that have been fully-aged. The so-enhanced fluxes, reaching up to ~ 1.9 MW/m2 motivate examination of such effects in large-scale, convective-boiling situations. Furthermore, the effect of coolant chemistry through molecular scale liquid-solid interactions at the contact line gives yet another evidence to further support the idea (Theofanous et al, 2002) that the basic mechanism of burnout is microlayer rupture by contact line instability. We have conducted a series of experiments on nanofluid boiling and nanofluid burnout. We discovered that presence of nanoparticles in the coolant alters both nucleation and nucleate boiling heat transfer behavior, making a fresh heater to behave like an aged one and significantly improving the heater’s resilience to burnout. We measured contact line motions for clean water and nanofluid and investigated mechanisms by which non-hydrodynamic factors govern the burnout. A basic understanding derived from our experiments and theoretical analysis led us to suggest that molecular wicking and nanoscale forces, as manifested in the effect of coolant chemistry and nanoparticles, offer new avenues to

Task Progress:	control and enhance resistance to burnout in pool boiling as well as in flow boiling and spray cooling (Theofanous and Dinh, 2006). The understanding that emerged in this work could also elucidate film dryout, i.e. burnout in high quality boiling as is of interest to NASA for Rankin cycle effort. As a follow-up of a suggestion made in Theofanous et al (2002) that high heat flux pool boiling can be beneficially observed in a thin-film geometry, BETA-B experimental setting was developed and BETA-B test campaign was conducted (Dinh and Tu, 2007). We used high-speed video and infrared imaging to study micro-hydrodynamics of evaporating liquid film. The data confirm that low-temperature (dark) spots in the heater's infrared images are bubble footprints. The video images suggest that the wall liquid film is tens of micrometers thick. It can be seen that bubbles nucleate and grow in the liquid film, with a thin liquid layer in the bubble's base and a thicker liquid ring at the bubble's edge. The bubble base's maximum size varies from 500 to 2000 μm. As the bubble expands into the upper vapor space, the liquid film (bubble dome) becomes thinner and eventually breaks off at the end of the bubble life. Following the bubble dome breach, the peripheral liquid spreads back into the bubble base area. Notably, bubble collective dynamics and pattern appears more chaotic and complex than that of thermal footprints, which are somewhat attenuated by the heater substrate thermal capacity. Analysis of the BETA data and processes suggests a physical picture of high heat-flux boiling that qualitatively includes (i) evaporation in the wall fluid layer is non-equilibrium in the sense that that bubble nucleation/expansion-induced agitation of the fluid layer leads to superheated interfaces; (ii) hot spots formed in the bubble base area represent superheated liquid layer; (iii) liquid spreading into the bubble base area upon the bubble collapse is driven by capillarity and expansion of bubbles in the vicinity; (iv) burnout occurring in irreversible hot spots is resulted from the failure of liquid spreading into the bubble base (hot spot) area to compensate for intense evaporation and dryout of the bubble base liquid layer. The BETA-B data also pave the way to explain several puzzles in boiling heat transfer, namely the relation between CHF and nucleation site density, the role of dynamic interactions between nucleating bubbles, and the effect of surface wettability on heater resilience to burnout. Most importantly, the new understanding of micro-hydrodynamics shall apply to high heat-flux boiling in pool boiling, flow boiling and spray cooling. Practical application of the research results: This project created a novel understanding of Boiling Crisis phenomenon in pool boiling. The scales separation phenomena discovered in BETA experiments and now elucidated in BETA-B experiments, open up a physics-based mechanistic modeling of high heat flux boiling and burnout. We can then translate this understanding to a means for quantitative prediction of critical heat flux, as a function of local (near-wall) two-phase flow conditions, appropriate and convenient for interrogation (modelling) in the frame of detailed multiphase CFD simulations. Furthermore, the thus-gained understanding and experience prove useful as a guidance to developing technologies to enhance the coolability limits in both pool and forced convection flows by manipulating surface characteristics, and flow geometries. Notably, in the BETA-B test series, aged heaters (with nanoporous surfaces) exhibited excellent resilience to burnout, with CHF on liquid-film boiling reaching to nearly 3 MW/m2. The result suggests that nanotreatment of the heater surface presents a very effective avenue to boiling enhancement. The NASA work has been leveraged with a contract from the Department of Energy concerning the coolability limits of nuclear reactor vessels that in case of severe accidents are immersed in water pools. In turn the results of this fundamentally oriented work for NASA had a tremendous influence and impact on the large scale experiments we run for this nuclear energy application, and helped resolve this practical issue.

Bibliography Type:	Description: (Last Updated: 10/17/2007)
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